Atish Mondi

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Education

Lovely Professional University — Phagwara, Punjab
B.Tech in Computer Science and Engineering — Graduated
CGPA: 7.06/10

Selected Projects

FL-Privacy-IDS

Python, PyTorch, Opacus, scikit-learn, Streamlit, Plotly

- Simulated **20 clients over 150 rounds** on UNSW-NB15 using FedAvg, FedProx, FedOpt, and FedNova across IID and Dirichlet non-IID splits.
- Created a Streamlit/Plotly explorer for saved accuracy and F1 curves, and integrated experimental Opacus DP-SGD to investigate privacy-utility trade-offs.

Purchase-to-Pay Conformance

Python, event logs, HTML/CSS, JavaScript

- Built a Python checker for five purchasing rules, distinguishing compliant, violating, and unknown cases from event order and cumulative quantities.
- Analyzed **251,734** BPI Challenge 2019 cases and built a case-inspection dashboard, ERP export validator, and **66-test** regression suite.
- Added timestamp, invoice-link, and source-hash validation so dashboard builds reject results that disagree with their event logs.

Rental Architecture and Recovery

Python, SQL, SQLite, worker processes

- Compared coupled, modular, and separate-worker rental architectures in Python/SQLite to evaluate booking availability and invoice consistency.
- Implemented outbox delivery, idempotent retries, and versioned invoice events, passing all **36 applicable synthetic scenario runs**.
- Built a paused migration workflow with exact invoice-row verification and rollback before booking writes resume.

Elbow ROM Video Analysis

Python, FastAPI, YOLO11m Pose, OpenCV, ReportLab

- Developed a FastAPI application for video-based elbow-angle analysis using pretrained YOLO11m Pose, OpenCV, and pixel-space geometry.
- Added mirrored-prediction agreement, confidence filters, frame inspection, and PDF/CSV/video exports, with **48 automated tests** covering core workflows.

Technical Skills

Languages: Python, SQL, JavaScript, C++, HTML/CSS

Frameworks: FastAPI, Flask, Streamlit, React

Data and ML: PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, OpenCV

Databases and Tools: SQLite, MySQL, Git/GitHub

Certifications

Google UX Design Professional Certificate: Google | Sep 2025

Cloud Computing: NPTEL | Oct 2024

Data Structures and Algorithms - Self-Paced (C++): GeeksforGeeks | Jul 2024

Generative AI with Large Language Models: DeepLearning.AI / AWS (Coursera) | Jun 2023